

M Abdullah Khan

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[IEEE Collabratec](#) | [Google Scholar](#) | [ResearchGate](#) | [Portfolio](#) | [Dhaka, Bangladesh](#)

OBJECTIVE

Results-driven Electrical and Electronic Engineering professional specializing in Robotics and Embedded Systems, with strong expertise in IoT, space technology, and industrial automation. Demonstrates proven capability in leading multidisciplinary engineering projects, contributing to internationally recognized research, and delivering scalable, innovation-driven solutions. Committed to advancing next-generation automation and space systems through applied engineering, data-driven design, and high-impact problem-solving.

EXPERIENCE

- Space Generation Advisory Council** Nov 2023 - Present
Research Assistant & NPoC (Bangladesh) Vienna, Austria
 - Coordinated national-level space initiatives and represented Bangladesh in global space advocacy.
 - Spearheaded to three multidisciplinary projects with international teams, driving cross-domain innovation till date.
 - Organized technical workshops and knowledge-sharing sessions to promote awareness of space and automation technologies.
- Ektukhani Robotics** Jan 2021 - Dec 2022
Content Writer & Coordinator Dhaka, Bangladesh
 - Developed engaging technical content on basic robotics, IoT & electronics for the company's website and publications.
 - Executed research to ensure accuracy and clarity in articles, increasing reader engagement by 30%.

EDUCATION

- United International University** Jan 2020 - Present
BSc. in Electrical & Electronic Engineering Dhaka, Bangladesh
 - Major: Robotics & Embedded Systems
- Dr. Mahbubur Rahman Mollah College** Aug 2017 - Jun 2019
Higher Secondary Certificate Dhaka, Bangladesh
 - Grade: GPA 4.67 (Out of 5.00)
- Mohanagar Ideal School & College** Jan 2015 - Mar 2017
Secondary School Certificate Dhaka, Bangladesh
 - Grade: GPA 5.00 (Out of 5.00)

PROJECTS

- Wearable Maternal Observatory Monitoring System** Feb 2025 - Aug 2025
Tools: Arduino, IoT sensors, Embedded C, Cloud-based monitoring platforms, Wearable technology [\[in\]](#)
 - Directed a 10-member team through the full development lifecycle of a wearable IoT health system.
 - Designed a wearable system to monitor maternal health in real-time using IoT-enabled sensors and embedded technologies.
 - Secured a Top 10 finish at the IEEE YESIST12 finals at Universiti Kebangsaan Malaysia.
- UIU CanSat: A Can-Sized Satellite** Nov 2023 - Jul 2025
Tools: Teensy, Sensors (Pressure, Temperature, Accelerometer, GPS), Telemetry software [\[v\]](#)
 - Architected and led the development of electrical systems and communication modules.
 - Developed a functional CanSat prototype for atmospheric data collection and telemetry.
 - Achieved 11th place globally in 2024 and 7th place in 2025 in the CanSat Competition, organized by the American Astronautical Society and sponsored by the US Naval Research Lab and NASA.
- UIU Mars Rover: Prototype for Planetary Exploration** Oct 2021 - Aug 2022
Tools: Arduino, C++, PCB design & soldering, System Integration, Gantt Chart management [\[v\]](#)
 - Led a cross-functional team, controlled project finances, and authored comprehensive project reports.
 - Contributed to the design and development of the electrical and mechanical systems.
 - Achieved 13th place globally and ranked highest in Asia at the University Rover Challenge 2022, organized by The Mars Society.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION, P=PRE-PRINT

- [S.1] R. E. Soglo, G. Filimónova, R. D. Sharma, P. Miralles, **M. A. Khan**, G. Furtado, J. Mohanty, M. Gupta, and H. Hall. **"Assessing the Environmental Consequences of Satellite Reentry: Impact on Ozone Layer Depletion Using the GEOS-Chem Global Chemistry-Transport Model"**, in 77th *International Astronautical Congress*, International Astronautical Federation, 2026.
- [J.1] Chakraborty, S., Fatima, A., Truong, A., Larini, R., **Khan, M. A.**, Tiwari, M., Ravikumar, B., Morales Ramirez, C. D., Jain, V., Khan, H. S., Lacoss, I., Wadhwa, P., Timmermans, R., Chu, J., & Agrawal, R. (2025). **AI-Powered Space Mission Risk Prediction: Real-Time Failure Detection and Adaptive Contingency Planning**. In 58th *IAA Symposium on Safety, Quality and Knowledge Management in Space Activities* (pp. 206–218). International Astronautical Federation (IAF). 58th IAA Symposium on Safety, Quality and Knowledge Management in Space Activities, Held at the 76th International Astronautical Congress (IAC 2025).
- [J.2] Yuvaraju, G., Morales Ramirez, C. D., Puleo, G., Suresh, S., Shekhar, S., Bello, A., Santhosh, V., Sumanta, S., Osadugba, A., **Khan, M. A.**, Taherkhany, M., & Habteslasie, E. (2025). **Lunar Vitalis: Unlocking Resources for Sustained Exploration**. In *IAF Space Exploration Symposium* (pp. 597–611). International Astronautical Federation (IAF). IAF Space Exploration Symposium, Held at the 76th International Astronautical Congress.
- [C.1] **Khan, M. A.. "Design and Performance Analysis of a 28 GHz Hexa-slot Antenna for Bangladesh's Emerging Smart Infrastructure in Medical Applications."** In *IEEE CS Bangladesh Chapter Symposium*, Vol. 3, Nov 22-23, 2024, Dhaka, Bangladesh. (2024).
- [C.2] **Khan, M. A.** (2023). **"The Effects of Blue Light Emission from Electronic Devices on Human Sleep Patterns and Health."** In *Research Journal of Advanced Engineering and Science*, 8.2 (2023): 33-35.
- [P.1] **Khan, M. A.**, Shamsuzzaman, M., Labib, F., Islam, S., Mashira, M., & Al-Saad, Md. A. (2023). **Bangladesh's Sustainable Water Production Solutions: An Analysis of Different Methods**. *Institute of Electrical and Electronics Engineers (IEEE)*.

SKILLS

- **Programming Languages:** Embedded C, C++, Python, Arduino, Verilog/HDL, Machine Learning, ROS.
- **Circuit Design & Simulation:** PCB Design, Circuit assembling, Multisim, Proteus, Matlab, Simulink.
- **Wireless Sensor Networks & IoT:** Zigbee, LoRa, Bluetooth Low Energy, IoT Systems, MQTT, ESP, Teensy.
- **Web Technologies & Content Creation:** HTML, CSS, JavaScript, Web Design, Adobe Photoshop, Adobe Illustrator, Content Writing & SEO Optimization
- **Database & Version Control:** MySQL, Git, GitHub, SPSS, Google Cloud, Network Security
- **Productivity & Communication Tools:** MS Office, Google Workspace, Slack, MS Project, Trello, Zoom
- **Research Skills:** Experimental Design, Data Analysis, Literature Review, Technical Writing, Project Documentation, Scientific Reporting

HONORS AND AWARDS

- **EEE DAY 2024 Champion** Nov 2024
Dept of EEE, United International University
 - Recognized as the Champion among 15 teams in a Research Poster Presentation contest, part of EEE Day 2024, hosted by the Department of Electrical & Electronic Engineering, United International University.
- **IEEE Region 10 WIE Pre-University Champion** Dec 2023
IEEE Region 10 Women in Engineering
 - Recognized for outstanding contribution and participation in IEEE Region 10 Women in Engineering pre-university initiatives in the year 2023.
- **IEEE R10 Ethics Champion** Oct 2023
IEEE Region 10 Educational Activities Committee
 - Awarded for excellence in promoting ethical standards and professional conduct in IEEE educational activities among IEEE student and professional communities.
- **Mujib 100 Idea Contest Finalist** Nov 2021
University Grant Commission (UGC), Bangladesh
 - Selected as one of the 40 finalists in the national "Mujib 100 Idea Contest," organized on the 100th Birth Anniversary of Bangabandhu Sheikh Mujibur Rahman.
- **EEE Day 2021 2nd Runner-up** Oct 2021
Dept. of EEE, Green University of Bangladesh
 - Achieved 2nd runner-up among 50 teams from different universities in a virtual poster presentation contest, part of EEE Day 2021, hosted by the Department of Electrical & Electronic Engineering, Green University of Bangladesh.

LEADERSHIP EXPERIENCE

- Chairperson, IEEE UIU Student Branch

Mar 2026 – Present

United International University

- Led the organization in hosting engineering workshops, seminars, and events, fostering a culture of scientific and research inquiry among students.
 - Served as Vice-Chair (Technical) (Mar 2025–Mar 2026), contributing to technical program development and execution.
 - Skillset: Leadership, Event Planning & Execution, Team Management, Budget handling, communications.
- Marketing Lead, IEEE Day

Feb 2025 – Jan 2026

Virtually, Dhaka, Bangladesh

- Launched and scaled social media campaigns, boosting engagement and participation across platforms.
 - Previous role: Communications Lead (Feb 2024 – Feb 2025) – Directed communication strategies, managing promotional efforts and boosting event participation, while collaborating with regional committees for smooth flow.
 - Skillset: Google Analytics, Meta Business Suite, MS Office.
- President, UIU Science Forum

Jan 2023 – Dec 2023

United International University

- Managed a core team of 15+ members and 50+ general members, ensuring efficient coordination and successful execution of club activities within the university campus.
 - Skillset: Leadership, Event Planning & Execution, Team Management.
- Joint Secretary (Public Relations), UIU Robotics Club

Jul 2022 – Jun 2023

United International University

- Orchestrated club competitions and robotics workshops, growing membership to 300+ participants for the CSE project show at my university.
 - Skillset: Event Coordination, Public Relations, Project Support.

VOLUNTEER EXPERIENCE

- Ambassador, IEEE Bangladesh Section Conference

Aug – Dec 2024

IEEE Bangladesh Section

- Championed IEEE at national conferences, driving participant engagement and institutional coordination.
- Volunteer in multiple roles

2021 – 2025

IEEE (Institute of Electrical and Electronics Engineers)

- Publicity Ambassador & Pilot, IEEE YESIST12 | Social Media Team, IEEEExtreme | PES & WIE DAY Ambassador.
- Public Relations Team, IEEE Computer Society BDC – Team SPARK

Apr – Dec 2024

IEEE Computer Society Bangladesh Chapter

- Managed communications and event promotions for workshops and technical competitions.
- Content Writing Coordinator, IEEE Computer Society BDC Web & IT Committee

May – Dec 2024

IEEE Computer Society Bangladesh Chapter

- Developed technical content for web platforms and IT initiatives, improving knowledge dissemination.

PROFESSIONAL MEMBERSHIPS

- American Spaces Bangladesh, US Embassy Dhaka. Membership ID: AC0001664

Sep 2024 - Present
- International Association of Engineers (IAENG), China. Membership ID: 264282

Oct 2021 - Present
- IEEE Member & Brand Ambassador, Dhaka, Bangladesh. Membership ID: 97205094

Sep 2020 - Present

CERTIFICATIONS

- Innovation Through Design: Think, Make, Break, Repeat - University of Sydney (Online Course)

Jun 2025
- Energy Harvesting - Sungkyunkwan University (Online Course)

Mar 2022
- Quick Start to IoT - RoboDoc Limited (Hands-on Course)

Jan 2022
- Robotics: Basic to Advanced Course - RoboTech Valley (Hands-on Course)

Apr 2021
- Introduction to the IoT and Embedded Systems - University of California, Irvine (Online Course)

Jun 2020

ADDITIONAL INFORMATION

Languages: Bengali (Native), English (Advanced Reading & Writing, Intermediate Listening & Speaking)
Research Interests: Robotics and Autonomous Systems, Space Exploration Technologies, Internet of Things (IoT), Embedded Systems, Industrial Automation and Control Systems

REFERENCES

- Dr. Md. Iqbal Bahar Chowdhury

Professor, Dept. of EEE

United International University

Email: iqbal@eee.uiu.ac.bd

Relationship: Capstone Supervisor
- Dr. Raqibul Mostafa

Professor, Dept. of EEE & Former Dean, SoSE

United International University

Email: rmostafa@eee.uiu.ac.bd

Relationship: IEEE Advisor & Faculty